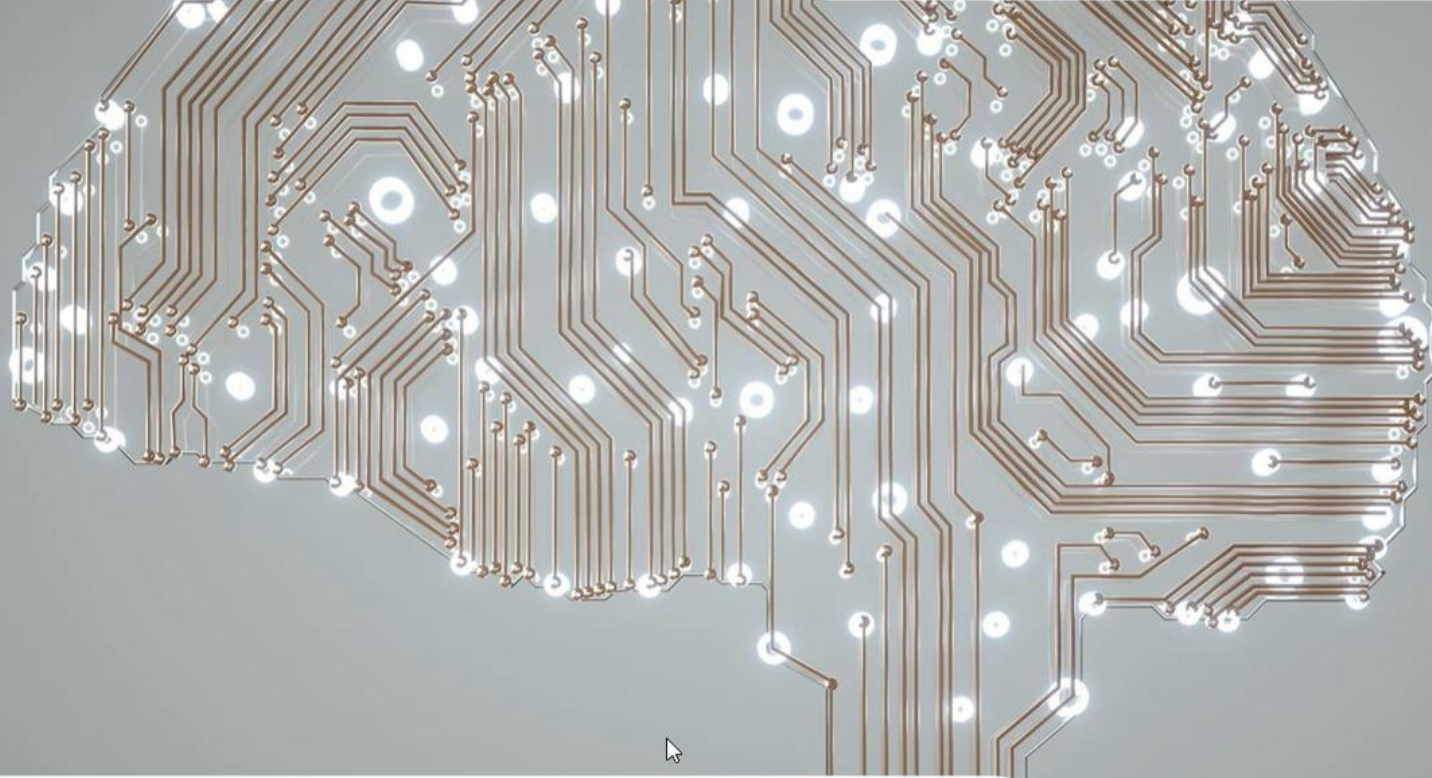
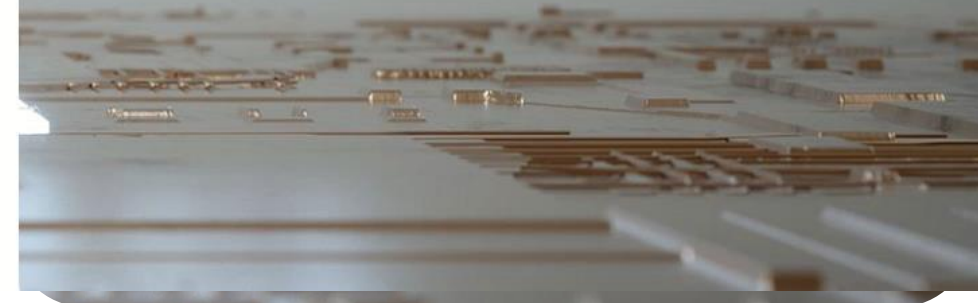




My AI Building Capstone:

Project, Process, and Products Story

Documenting my AI learning and iterating journey



AI Builder Bootcamp Cohort 8 Material Chatbot

GOAL

DESCRIPTION

Learning Memorialization

Transcribe and make slide decks to be recall-ready Winter 2026 learning

Use Case "Why"

So I can have an interactive personal repository for AI builder foundations taught and learned

End-to-End Workflow for Slide Deck Chatbot

STEPS:	DESCRIPTION
Slide Deck PDFs from ABB Cohort 8 Maven website	Downloaded all lessons' PDFs from Sessions 1 through 6 and compiled them (304 slides)
PDF Parsing <i>(learning from Session 1)</i>	PyMuPDF extracts text per slide page, keeping slide number and title as metadata
Hybrid RAG Chunking <i>(learning from Session 3)</i>	Each slide becomes a TextNode; long slides are sub-chunked via SentenceSplitter
Dense Retrieval <i>(learning from Session 3)</i>	OpenAIEmbedding + VectorStoreIndex, persisted for efficient access
Sparse Retrieval <i>(just learned now)</i>	BM25Retriever for keyword matching, rebuilt on each load
Fusion <i>(just learned now)</i>	QueryFusionRetriever with Reciprocal Rank Fusion (top-6 slides)
Generation <i>(learning from Session 2)</i>	gpt-4o-mini with conversation history and retrieved context
Develop UI through Streamlit <i>(learning from Sessions 1 & 6)</i>	Streamlit chat with slide-number attribution and sidebar navigation

Turning chunks into meaning

Perform keyword search and relevance

Multiple searches to rank & merge the top 6 most relevant slides

Local and Online Chatbot Versions

VERSION

DESCRIPTION

Local

Runs locally with persisted vector index for fast, secure access: [Proj6 Capstone Chatbot.py](#)

Online

Work in Progress ...

Hosted version for broader access and collaboration : [proj6_streamlit_ui.py](#)
It has a Google Drive link with MCP server

Next Steps and Continued Learning



ACTION

PURPOSE

Enjoy as Digital Souvenir

Personal repository for AI learning

Continue Learning

Add and retrieve more material

Organize Future Learning

Structure and manage ongoing education

AI Evaluation

Refine prompts and improve chatbot performance